Lubrication Service — Centrifugal Pump

Lubrication Issue — Stage 2 Broadband Pattern

Assets: AST_PMP_001 (Cooling Pump A), AST_PMP_002 (Cooling Pump B) | LINE_002

Bearing Type: SKF6208 | Est. Duration: 3 h | Crew: 1–2 | Priority: MEDIUM

SOP_003 | v1.1 | Revised: 2026-03-15

1. Purpose and Scope

This procedure covers the oil drain, inspection, and refill service for centrifugal pump bearings where the Failure Intelligence Agent has classified a lubrication_issue — Stage 2. Stage 2 lubrication issue presents as broadband vibration increase (1.3–2.0× baseline) and temperature rise (10–20°C above baseline), WITHOUT a dominant single frequency band and WITHOUT kurtosis exceeding 5.0. This pattern indicates oil degradation or insufficient oil volume, not bearing surface damage. First action is always oil service + 24h recheck. If kurtosis exceeds 5.0 after service, escalate to bearing replacement. Applies to SKF6208 bearings on AST_PMP_001 and AST_PMP_002 on LINE_002.

2. Prerequisites

- Failure Intelligence diagnosis confirmed: lubrication_issue, Stage 2 — NOT outer_race_fault
- Kurtosis at detection: below 5.0 (if above, escalate to bearing replacement, not lubrication service)
- PART_005 (Industrial Oil ISO68) available in BIN_L2 — minimum 200 mL
- LOTO_002 current (AST_PMP_001) or LOTO_006 current (AST_PMP_002)
- Operations context confirms pump can be taken offline: LINE_002 contingency LINE_001
- Oil sampling kit for drain oil condition assessment
- Approved 3-hour maintenance window (CTX_002 or CTX_010)

3. Safety Requirements

3.1 LOTO Isolation Procedure

Permit — AST_PMP_001	LOTO_002 | Isolation: Valve-7, PRV-02
Permit — AST_PMP_002	LOTO_006 | Isolation: Valve-8, PRV-03
Energy Types	Hydraulic, Electrical
Isolation Steps	Isolate hydraulic supply at valve. Open PRV to release line pressure. De-energise motor at MCC. Apply padlocks. Verify zero pressure at gauge.
Verification	Confirm zero pressure on line gauge. Confirm motor does not start. Confirm no hydraulic flow at drain point.

3.2 PPE and Zone Requirements

Mandatory PPE	Chemical splash goggles, nitrile gloves, face shield for oil drain step, safety boots
Zone — AST_PMP_001	B — Unrestricted approach; standard PPE sufficient
Zone — AST_PMP_002	A — Normal approach; standard PPE applies
Oil Spill Risk	Place drip tray under bearing housing before opening drain plug

4. Tools and Materials Required

Part Number	Description	Quantity	Bin Location
PART_005	Industrial Oil ISO68 (Castrol / equivalent)	200 mL	BIN_L2

- Oil drain container (minimum 500 mL)
- Oil sample vial and label for drain oil condition record
- Oil fill syringe or squeeze bottle — 200 mL capacity
- Infrared thermometer
- Portable vibration meter
- Clean lint-free rags
- Torque screwdriver for drain/fill plugs

5. Step-by-Step Procedure

1	Confirm diagnosis is lubrication_issue (Stage 2) and kurtosis is below 5.0 before starting.
2	Apply LOTO per Section 3.1. Verify zero hydraulic pressure and zero electrical energy.
3	Position drip tray under bearing housing.
4	Remove drain plug from bearing housing. Drain oil completely into drain container.
5	Collect 5 mL oil sample in labelled vial for condition record: note colour, smell, particle content, and viscosity impression.
6	Inspect drained oil: clear amber = normal degradation. Dark or black = heavy oxidation. Metallic particles = bearing surface damage — STOP, escalate to bearing replacement (SOP not applicable).
7	If oil is acceptable (amber, light oxidation only): continue with refill. If metallic contamination found: STOP, escalate.
8	Refit drain plug. Torque to 12 Nm.
9	Fill bearing housing with 200 mL fresh PART_005 (Industrial Oil ISO68) via fill port.
10	Refit fill plug. Torque to 12 Nm.
11	Remove LOTO. Controlled restart: bring pump to operating speed.
12	Take vibration reading at 30 minutes. Note: improvement after oil service is not immediate — 30-minute reading is early warning only.
13	Run at full operating condition for 24 hours before formal QA check.
14	Take final vibration, temperature, and kurtosis readings at 24 hours. Compare against Section 7 criteria.
15	Update bearing_master.last_lubrication_date to today. Record oil condition findings on work order.
16	Close work order. If 24-hour readings have not returned to baseline, schedule bearing inspection.

6. Critical Specifications

Oil Type	PART_005 — Industrial Oil ISO68 (Castrol Hyspin or equivalent)
Oil Volume	200 mL per bearing housing — do not overfill
Re-lube Interval	Every 60 days. NOTE: original system setting of 90 days was incorrect — 60 days confirmed correct for this operating temperature (CASE_002 correction).
Drain Plug Torque	12 Nm
Oil Condition Rejection	Metallic particles visible, burnt smell, or viscosity clearly reduced → bearing damage; escalate to SOP for replacement

7. Post-Service Acceptance Criteria (at 24 hours)

The 30-minute reading is for trend monitoring only. PASS/FAIL is determined at 24 hours.

Parameter	AST_PMP_001 (BRG_005/006)	AST_PMP_002 (BRG_007/008)	Failure Action
Vibration RMS at 24 h	≤ 3.5 mm/s	≤ 3.0 mm/s	Schedule bearing inspection

Parameter	AST_PMP_001 (BRG_005/006)	AST_PMP_002 (BRG_007/008)	Failure Action
Bearing Temperature at 24 h	≤ 65°C	≤ 60°C	Check oil fill level; re-inspect
Kurtosis at 24 h	≤ 3.5	≤ 3.0	Escalate: bearing damage suspected
Signal Quality Score	≥ 0.97	≥ 0.97	Check CH_005/006 or CH_007/008

8. Common Issues and Troubleshooting

- If vibration does not improve at 24 h (remains above 3.5× baseline) → bearing damage is present; lubrication alone cannot resolve. Escalate to bearing replacement.
- If temperature rises above 70°C at 30 min → possible overfill; drain to mark and restart.
- If metallic particles found in drained oil → STOP. Do not refill. Escalate immediately to bearing inspection and likely replacement.
- If kurtosis remains above 4.0 at 24 h → cage fault developing; schedule bearing replacement within 5–10 days.
- Re-lube interval must be 60 days, not 90 days — system parameter should be updated at work order close (CASE_002 root cause).

9. Related Documents

- SOP_006 — Post-Repair QA Procedure (all assets)
- LOTO_002 permit record — AST_PMP_001
- LOTO_006 permit record — AST_PMP_002
- CASE_002 — Lubrication issue, AST_PMP_001, lube interval root cause confirmed
- CASE_009 — Preventive lube check, AST_PMP_001, BRG_006 (NDE), no fault
- CASE_010 — Vibration inspection, AST_PMP_002, BRG_008 (NDE), healthy